

Landau's thesis

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In his Ph.D. thesis, Edmund Landau presented an elementary proof of the fact that $\theta(x) \sim x$ implies

$$\sum_{n=1}^{\infty} \frac{\mu(n)}{n} = 0, \quad (1)$$

where $\theta(x) = \sum_{p \leq x} \log p$ is the Chebyshev function. This closed form was conjectured by Euler and first proved by von Mangoldt. It can be shown with a little effort that $\theta(x) \sim x$ implies $M(x) = o(x)$. Using the second estimate we will prove (1). The converse follows simply by partial summation as

$$\sum_{n \leq x} \mu(n) = x \sum_{n \leq x} \frac{\mu(n)}{n} - \int_1^x \left(\sum_{n \leq u} \frac{\mu(n)}{n} \right) du = o(x)$$

since $\int_1^x f(u) du = o(x)$ for every function f with $f(x) = o(1)$.

We remark here that one cannot obtain (1) just by an application of partial summation formula since doing so leads to

$$\sum_{n \leq x} \frac{\mu(n)}{n} = \frac{M(x)}{x} + \int_1^x \frac{M(u)}{u^2} du.$$

Although the first term to the right tends to 0 the same cannot be said about the integral. A priori the integral is only $o(\log x)$.

Theorem 1. $M(x) = o(x)$ implies (1).

Proof. For $x \geq 1$ we have the identity

$$1 = x \sum_{n \leq x} \frac{\mu(n)}{n} - \sum_{n \leq x} \mu(n) \left\{ \frac{x}{n} \right\}.$$

Due to this it suffices to show that $\sum_{n \leq x} \mu(n) \{x/n\} = o(x)$. For a positive integer N we have

$$\begin{aligned} \sum_{x/N < n \leq x} \mu(n) \left\{ \frac{x}{n} \right\} &= \sum_{k=1}^{N-1} \sum_{x/(k+1) < n \leq x/k} \mu(n) \left(\frac{x}{n} - k \right) \\ &= x \sum_{x/N < n \leq x} \frac{\mu(n)}{n} - \sum_{k=1}^{N-1} k (M(x/k) - M(x/(k+1))) \end{aligned}$$

Let S_1 and S_2 denote the first and the second sum respectively. We will show that $S_1 = o(1)$ and $S_2 = o(x)$ for some choice of N . By partial summation we have

$$S_1 = \frac{M(x)}{x} - \frac{M(x/N)}{x/N} + \int_{x/N}^x \frac{M(u)}{u^2} du.$$

Let $E(x) = \sup_{y \geq x} |M(y)|/y$. Then we have $E(x) = o(1)$. The integral to the right is $\leq E(x/N) \log N$ and hence $S_1 \ll E(x) + E(x/N) + E(x/N) \log N$.

As for S_2 we note that

$$S_2 \ll x \sum_{k \leq N} \frac{|M(x/k)|}{x/k} \ll xNE(x/N).$$

Thus we find that

$$\sum_{x/N < n \leq x} \mu(n) \left\{ \frac{x}{n} \right\} \ll xE(x) + xE(x/N) + xNE(x/N) \ll xNE(x/N).$$

Let $N = \lfloor \min(x^{1/2}, E(x^{1/2})^{-1/2}) \rfloor$. Then $E(x/N) \leq E(x^{1/2})$ as E is monotonically decreasing. Now we have $NE(x/N) \leq NE(x^{1/2}) \leq E(x^{1/2})^{1/2}$. Thus we conclude that

$$\sum_{x/N < n \leq x} \mu(n) \left\{ \frac{x}{n} \right\} = o(x).$$

Finally, we can bound the partial sum trivially as

$$\sum_{n < x/N} \mu(n) \left\{ \frac{x}{n} \right\} \ll \frac{x}{N}$$

and so the sum is $o(x)$ for the above choice of N . This completes the proof. $\square$